

CAN Bus messages

CAN Bus messages

CAN Bus on Hyundai Tiburon operates at a speed of 500kb/s and so do the SIMK4x ECUs.

SIMK4x reads the following messages for operation, if the corresponding module is present in car:

- ARBID 0x153 ASC1 Traction Control
- ARBID 0x43F EGS1 Automatic Transmission

SIMK4x sends the following messages:

- ARBID 0x2A0 DME5 Engine Control Unit (since ca663056)
- ARBID 0x545 DME4 Engine Control Unit
- ARBID 0x329 DME2 Engine Control Unit
- ARBID 0x316 DME1 Engine Control Unit

In addition to that, there are messages that are not used by the ECU, but still present on the CAN Bus:

- ARBID 0x1F0 ASC2 Traction Control

DBC file of Tiburon's CAN Bus: Tiburon_SIMK4x.dbc [mirror]

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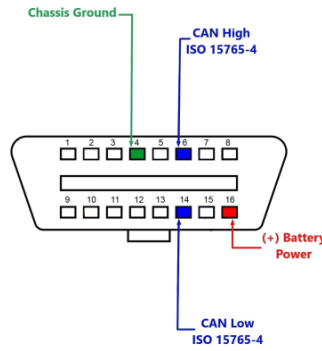


Example of a candump log imported into SavvyCAN with Tiburon_SIMK4x.dbc file loaded

Connection points

There is only one CAN bus found in the Tiburons, there are no gateways present.

CAN Bus is exposed on the standard pins in the OBD2 connector as in the following image (vehicle side):



Tiburon's OBD2 plug pinout, limited to power and canbus

Depending on vehicle's production date and trim, there are between 1 to 4 modules on the bus.

Modules connected to the CAN Bus

Module name	Connector name		CAN high pin	CAN low pin	CAN ID
ECM	2.0 pinout	2.7 pinout	see pinout	see pinout	0x545, 0x329, 0x316
TCM	2.0 pinout	2.7 pinout	see pinout	see pinout	0x43F
Multigauge unit (if equipped)	M42		7	1	
ABS module (if equipped)	E37		11	10	0x153

Messages

DME1 0x316

Refresh Rate: 10ms

- **Byte 0 - bitfield**
 - Bit 0 - SWI_IGK_CAN: Terminal 15 –KEY ON
 - Bit 1 - F_N_ENG_CAN: Error – engine speed signal
 - Bit 2 - ACK_TCS_CAN (LV_ACK_TCS): Acknowledgement of TCS. This bit is set to 0 if no CAN messages from the TCS were received for at least 500ms
 - Bit 3 - PUC_STAT: Engine in fuel cut off (LV_PUC - Trailing Throttle Fuel Cut Off)
 - Bit 4 - TQ_COR_STAT_CAN: Status of torque intervention 0 - The desired intervention regarding ignition angle retardation and cylinder shut-off is executed. (Default value) 1 - The desired intervention regarding ignition angle retardation and cylinder shut-off is executed; however, the requested target torque can not be adjusted precisely (torque steps) 2 - The torque reduction regarding the ignition angle retardation cannot be completely executed. A cylinder shut-off is not possible at this time. Therefore a remaining torque (as difference between TQI_ASR/GS_REQ and TQI_AV) is present and cannot be reduced. 3 - The desired torque intervention for TCS regarding the ignition angle and cylinder shut-off can no longer be executed. The torque intervention is terminated, the engine management system resets the requested engine torque to the TQI value using a ramp.
 - Bit 6 - RLY_AC_CAN: Activation, air conditioner compressor relay
 - Bit 7 - F_SUB_TQI_CAN: Error on torque measure or calculation
- **Byte 1 - TQI_TQR_CAN**
Indexed Engine Torque in % of C_TQ_STND (including ASR/MSR/ETCU/LIM/AMT/GEAR intervention)
Calculation = HEX * 0.39
- **Byte 2 - N_ENG [LSB]**
- **Byte 3 - N_ENG [MSB]**
Engine Speed in rpm
Calculation = ((HEX[MSB] * 256) + HEX[LSB]) * 0.15625
- **Byte 4 - TQI_CAN**
Indicated Engine Torque in % of C_TQ_STND (based on PVS, N, AMP, TIA ,TCO, IGA, PUC so ip_tqi_pvs__n__pvs)
Calculation = HEX * 0.39
- **Byte 5 - TQ_LOSS_CAN (TQFR_CAN)**
Engine Torque Loss (due to engine friction, AC compressor and electrical power consumption) in % of C_TQ_STND
Calculation = HEX * 0.39
- **Byte 6 - VS (VS_CAN)** Vehicle speed in km/h
- **Byte 7 - TQI_MAF_CAN**
Needs verification
Theoretical Engine Torque in % of C_TQ_STND after charge intervention (based on MAF & IGA so ip_tqi_maf__n__maf)
Calculation = HEX * 0.39

Example messages:

05 0B B8 0C 0B 0A 00 77

```
05 0C B8 27 0C 0B 46 80
```

```
0D 00 84 27 00 0B 31 82
```

DME2 0x329

Refresh Rate: 10ms

- **Byte 0 - Bitfield**
 - Bit 0 - CAN Version (CAN_VERS_CAN),
 - Bit 1 - TCU configuration (CONF_TCU_CAN)
 - Bits 2-3 - OBD freeze frame (OBD_FRF_ACK_CAN) @TODO: Not sure about the length
 - Bit 4-5 - Torque scaling factor (TQ_STND_CAN) (6 bits long) @TODO: Not sure about the length
 - Bit 6 - MUL_CODE: Identification of MUL_INFO
- **Byte 1 - TCO (TEMP_ENG_CAN)**

Engine (Coolant) Temperature in °C
Calculation = (HEX * 0.75) - 48°C
Init: 0xFF
Min: 0x01 (-48°C)
Max: 0xFF (142.5°C)
- **Byte 2 - AMP_CAN**

Ambient Pressure in hPa
Calculation = (HEX * 2) + 598hPa
Init = 0x00
Min = 0x01 (600hPa)
Max = 0xFE (1106hPa)
Error = 0xFF
- **Byte 3 - VB_OFF_ACT_CAN**

Detection of battery disconnection (0x10 - battery is connected)
- **Byte 4 - ACK_ES_CAN**

Acknowledgement „engine stopped“
- **Byte 5 - TPS_CAN**

Accelerator Pedal Position in % of PVS_MAX
Calculation = HEX * 0.390625
Init = 0x00
Min = 0x01 (0%)
Max = 0xFE (99.2%)
PVS Error = 0xFF
Values around 10-15% are to be expected for an un-touched pedal
- **Byte 6 - bitfield**
 - **Bits 0-4: ENG_CHR_CAN**

Engine Characteristic – Kind of fuel, ETS. 0xFF - gasoline
 - **Bits 5-7: Unused**
- **Byte 7 - Unused**

Example messages:

```
4F B2 82 10 00 20 FF
```

```
4F B0 82 10 00 32 FF
```

```
4F B1 84 10 00 51 FF
```

DME4 0x545

Refresh Rate: 10ms

- **Byte 0 - Bitfield**
 - Bit 0 - LV_LOCK_IMOB (IM_AUTHEN_CAN): Authentication "immobilizer" (0/1), 1 bit long. @TODO confirm: this value is likely 0 before successful immo authentication
 - Bit 1 - L_MIL_CAN: Check Engine Lamp or Malfunction Indication Lamp (MIL for CARB/OBDII) (0/1), 1 bit long
 - Bit 2 - CONF_IMOB (IM_STAT_CAN): Status "immobilizer" (0/1), 1 bit long. @TODO confirm: this value likely shows whether the immobilizer is enabled in the calibration zone
 - Bit 3 - AMP_CAN: Atmospheric Pressure [mmHg]. 5 bytes long
- **Byte 1 - FCO[LSB] (FCO_SUM_DIF)**
- **Byte 2 - FCO[MSB] (FCO_SUM_DIF)**

Fuel Consumption in microliters (µl). Formula: HEX * 0,128
- **Byte 3 - VB**

Battery voltage. Formula: HEX * 0.1
- **Byte 4 - Unused**
- **Byte 5 - Unused**
- **Byte 6 - Unused**
- **Byte 7 - Unused**

Example messages:

```
F6 61 01 89
EE FA 02 89
EE 8B 02 88
```

DME5 0x2A0

Refresh rate: 10ms

@TODO. The values below **have not been even looked at** on any Tiburon candump and instead come from a newer engine control unit. **Some might be accurate, or nothing may be accurate at all.** Feel free to verify or supply some CAN logs

- **Byte 0** - bitfield
 - Bit 0 - ECGPOvrd: Driver override (not relevant)
 - Bit 1 - QECACC: Failure in ACC-message detected by engine control (not relevant)
 - Bit 2 - ECFail: Function failure engine power control (not relevant)
 - Bit 3 - SwitchOffCondExt: ACC shut off condition detected (not relevant)
 - Bit 4 - BLCEFail: Failure brake light switch detected by engine control (not relevant) **extremely unlikely to be used, as the brake light switch is not connected to the ECU on GK platform**
 - Bit 5 - Unused
 - Bit 6 - Unused
 - Bit 7 - Unused
 - Bit 8 - Filtered Accelerator Pedal value(not relevant)
- **Byte 1** - TIA_CAN: Intake Air temperature Temperature in Celsius, formula: HEX * 0.75.
- **Byte 2** - bitfield
 - Bits 0-7: STATE_DC_OBD: Status of rate-based monitoring
 - Bit 8: INH_DC_RBM - Inhibition of rate based monitoring
- **Byte 3** - CTR_IG_CYC_OBD: Ignition cycle counter [LSB]
- **Byte 4** - CTR_IG_CYC_OBD: Ignition cycle counter [MSB]
- **Byte 5** - CTR_CDN_OBD: General denominator calculation [LSB]
- **Byte 6** - CTR_CDN_OBD: General denominator calculation [MSB]
- **Byte 7** - Unused
- **Byte 8** - Unused

Example logs

Here are some example canbus dumps. They were recorded on a 2008 Tiburon, 2.0 (cvvt) with manual transmission, fitted with ABS and ESP.

https://drive.google.com/drive/folders/1j9GmhrkH-Lm7k_E4v4JQ_gHB_lFeJ5_U?usp=sharing

2008_2.0_normal_city.log - This one features "eco-driving" around the city, without exceeding 40mph.

2008_2.0_normal_highway.log - eco-driving around the highway, without exceeding 80mph.

2008_2.0_spirited_highway.log - changing gears near the limiter until getting to the 110mph mark and mostly staying there for the whole duration of the log.

2008_2.0_spirited_city.log - dynamic stroll through the city. This is the only log during which ESP intervention happened - it's reflected in ESP's module CAN messages.

Rest of the logs should have self-explanatory names.

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